

Problem Statement #10 — Student Club Event Ticketing & Budget Portal

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1. Requirements Table

Actors: Club Lead, Faculty Coordinator, Finance Officer, Dean, Campus Admin, Attendee, Volunteer (check-in staff).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator -> Finance Office -> Dean).	High	Pass: Status moves to the next stage only after the current approver digitally signs off, in order. Fail: A request reaches the Dean without Finance Office sign-off.	Given in problem statement — ensures no proposal skips a required financial check before final approval.	
FR-002	Functional	The system shall allow a Club Lead to submit an event proposal containing event title, date, venue, expected attendance and an itemised budget.	High	Pass: Proposal is saved with a unique Proposal ID and status 'Submitted' only when all mandatory fields are filled in. Fail: Submission goes through with the budget field or event date left blank.	Nothing can be routed for approval until the club actually enters what it needs and how much it will cost.	
FR-003	Functional	Once a proposal is fully approved and registration opens, the system shall generate one unique QR-coded ticket per registered attendee.	High	Pass: Every ticket issued has a QR code that is unique across all events on the platform. Fail: Two different registrations (same or different events) are issued an identical QR code.	Duplicate or guessable codes at the gate would let people walk in without a valid registration.	
FR-004	Functional	The system shall give Campus Admin a single dashboard listing every club's events with their current approval status, budget and attendance figures.	Medium	Pass: A status change made by an approver is reflected on the Campus Admin dashboard within 5 seconds. Fail: Dashboard still shows the old status after 5 seconds.	Admin currently has no campus-wide view of club spending; this replaces manual follow-up emails with approvers.	
FR-005	Functional	The system shall email the Club Lead automatically whenever their proposal's approval status changes.	Medium	Pass: Email reaches the Club Lead's registered address within 2 minutes of the status change. Fail: No email is sent, or it	Club leads should not have to keep logging in and checking just to know where their request is stuck.	

				arrives more than 10 minutes late.		
NFR-001	Nonfunc.	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Median validation time exceeds 100 ms during peak load testing.	Given in problem statement — slow scans cause queues at the gate, especially for large events.	
NFR-002	Nonfunc.	The portal shall remain available at least 99.5% of the time during active event check-in windows, excluding pre-announced maintenance.	High	Pass: Uptime logs for a check-in window show availability of 99.5% or higher. Fail: Downtime during a check-in window exceeds 0.5% of that window's duration.	An outage right at check-in is the worst possible time — it stalls the entire queue at the venue.	